TEST 4—FORM CODE 1Z

SECTION 1 Time—30 minutes 40 Questions

For each question in this section, choose the best answer and fill in the corresponding oval on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. RECUPERATE: (A) concentrate (B) limit (C) sicken (D) seek help (E) change hands

2. FRUITLESS: (A) sour (B) adult (C) responsive (D) malodorous (E) productive

3. REBELLION: (A) ability to lead (B) desire for wealth (C) expectation of success (D) acquiescence to authority (E) conservation of resources

4. GIDDY: (A) well-meaning (B) undemanding (C) unpretentious (D) levelheaded (E) knowledgeable

5. ESTEEM: (A) modesty (B) curiosity (C) disrespect (D) feigned unconcern (E) restless uncertainty

6. UNFLAPPABLE: (A) talented (B) prosperous (C) sympathetic (D) easily upset (E) humbly grateful

7. REFUGE: (A) misrepresentation (B) ornamentation (C) offensive statement (D) optimistic attitude (E) dangerous place

8. PROSTRATION: (A) vitality (B) brevity (C) priority (D) fixity (E) unity

9. TERSE: (A) open-minded (B) long-winded (C) overpermissive (D) easily amused (E) tactlessly honest

10. VIABLE: (A) necessary (B) infamous (C) mesmerized (D) calculable (E) moribund

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that, when inserted in the sentence, best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. The women remained convinced that the adverse ruling was only a temporary setback—that their progress had merely been ----, not halted.

(A) enhanced (B) negated (C) interrupted (D) perpetuated (E) terminated

12. Science has conquered the forces of nature and compelled them to serve humanity; inhospitable forests have been ---- fertile fields.

(A) incorporated into
(B) transformed into
(C) developed from
(D) combined into
(E) produced by

13. It is unnecessary to say that their philosophy is ---- to his; for, since they knew little else besides his work, how could their philosophy be very ----.

(A) hostile..creative (B) identical..imitative (C) similar..different (D) indebted..parallel (E) akin..uniform

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14. By labeling their opponents' views a ---- doctrine, they persuaded themselves that it would fail to ---- through the years.

(A) reliable..survive
(B) foolish..succumb
(C) rigid..adapt
(D) sound..flourish
(E) hateful..regress

15. I find it hard to ---- such a ---- translation since it has much the same effect on Tourneau's lucid and vibrant prose that a steamroller would have on a flower bed.

(A) overrate..fluent
(B) praise..sublime
(C) excuse..pedestrian
(D) censure..ludicrous
(E) condone..deft

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep (B) anger : madness (C) smile : amusement (D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. FANG:SNAKE :: (A) pit:viper (B) tusk:elephant (C) hoof:horse (D) spot:leopard (E) wing:robin

17. COLOR : DULLED ::
(A) vision : scrutinized
(B) ore : mined
(C) tone : muted
(D) alcohol : preserved
(E) experiment : verified

18. THREAD : BUTTON :: (A) nail : hammer (B) alphabet : language (C) paste : wallpaper (D) handle : pitcher (E) frame : picture

19. EXPEL : PUPIL :: (A) acquit : defendant (B) deter : criminal (C) deport : alien (D) rebuke : child (E) exploit : friend

20. NAVIGABLE : WATERWAY ::
(A) inflatable : balloon
(B) inhabitable : home
(C) accessible : voyage
(D) invulnerable : fort
(E) passable : trail

21. GNAT : INSECT :: (A) bird : mammal (B) minnow : fish (C) peacock : eagle (D) python : venom (E) butterfly : moth

22. BEQUEST : HEIR :: (A) blood : donor (B) meal : waiter (C) thrift : miser (D) gift : recipient (E) cause : protester

23. DWINDLE : SIZE :: (A) melt : warmth (B) wither : vigor (C) gamble : luck (D) petrify : age (E) multiply : number

24. MURAL : VIGNETTE ::
(A) epic : ballad
(B) drama : script
(C) portrait : statue
(D) symphony : orchestra
(E) rhythm : syncopation

25. OMNIPOTENT : POWER :: (A) universal : deity (B) tyrannical : actions (C) eternal : duration (D) total : application (E) law : government

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Each passage below is followed by questions based on its content. Answer the questions following each passage on the basis of what is stated or implied in that passage.

For a long time, African religions were written off as only animism. We know now that they represent more, actually a very complicated system of philosophy. Like peoples everywhere, the Africans have wrestled with the big questions. What is the nature of existence? What happens after death? Is life a gigantic hoax or has it purpose and meaning?

The answers Africans have given to these questions have determined the form of their religions. There is, to begin with, a supreme God who created the earth. There is also a pantheon of lesser gods identified sometimes with terrestrial objects. Intertwined with these concepts are the cults of fate and ancestor worship. Undergirding all is the basic concept of "life forces." The life force of a creator is thought to be present in all things, animate and inanimate. This force, "a kind of individualized fragment of the Supreme Being itself," continues to exist, even after the death of the individual. It continues, the Africans say, in a pure and perfect state which could influence the lives of living things.

Some scholars have found parallels between African philosophy and modern physics. "African thought," they say, "is conditioned by African ontology, that is, their theory of the nature of being. For them being is a process and not a mere state, and the nature of things is thought of in terms of force or energy rather than matter; the forces of the spirit, human, animal, vegetable, and mineral worlds are all constantly influencing each other, and by a proper knowledge and use of them, human beings may influence their own lives and those of others."

Religion, to many Africans, is important in all aspects of life. Every event is suffused with religious significance, and the climax of life is death. The African attitude that death is a transition from one state to another, anthropologists say, survived the Atlantic crossing and took root in the soil of black existence in America.

26. It can be inferred from the passage that the form of African religions has been determined by

(A) a belief in the principle of individualism
(B) customs handed down from ancient civilizations
(C) a philosophy concerning life and death
(D) practices transmitted by spiritual leaders
(E) a desire for peace and tranquility

27. The author's attitude toward African religion is best described as

(A) facetious
(B) dubious
(C) censorious
(D) sympathetic
(E) indifferent

28. In terms of African philosophy, the word "god" could be applied to which of the following?

I. A creator
II. A force that animates living creatures
III. Energy within inanimate objects

(A) I only
(B) I and II only
(C) I and III only
(D) II and III only
(E) I, II, and III

29. The author is primarily concerned with discussing the

(A) basic philosophy underlying African religions
(B) efforts of scholars to understand African religions
(C) relationship of African religion to Western philosophy
(D) significance of African ritualism
(E) African's attitude toward a life after death

30. Which of the following statements is in accord with the principles of African ontology as presented in the passage?

(A) The spirit of humanity has no influence upon its destiny.
(B) The interaction of all spirits, animate and inanimate, influences humanity.
(C) Humanity is controlled by capricious gods.
(D) The spirit of humanity controls the spirit of the inanimate.
(E) The spirit of humanity should be removed from worldly considerations.

The reading passages in this test are brief excerpts or adaptations of excerpts from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

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1

The differences in relative growth of various areas of scientific research have several causes. Some of these causes are completely legitimate results of social needs. Others are a legitimate consequence of particular advances in science being to some extent self-accelerating. Some, however, are less legitimate processes of differential growth in which preconceptions of the form scientific theory ought to take, by persons in authority, act to alter the growth pattern of different areas. This is a new problem which is probably not yet ineradicable; but it is a frightening trend.

This trend began during the Second World War, when several governments came to the conclusion that the specific demands that a government may want to make of its scientific establishment cannot generally be foreseen in detail. It can be predicted, however, that from time to time questions will arise which will require specific scientific answers. It is therefore generally valuable to treat the scientific establishment as a resource or machine to be kept in functional order. This seems to be most effectively done by supporting a certain amount of research not related to immediate goals but of possible consequence in the future. This kind of support, like all government support, requires decisions about the appropriate recipients of funds. Decisions based on utility as opposed to lack of utility are straightforward. But a decision among projects none of which has immediate utility is more difficult. The goal of the supporting agencies is the laudable one of supporting "good" as opposed to "bad" science, but a valid determination is difficult to make. Generally, the idea of good science tends to become confounded with the capacity of the field in question to generate an elegant theory. However, the world is so made that elegant systems are in principle unable to deal with some of the world's more fascinating and delightful aspects. New forms of thought as well as new subjects for thought must arise in the future as they have in the past, giving rise to new standards of elegance.

The fact that science has become an adjunct of national policy does not, in itself, necessarily eliminate the possibility of new and radical concepts arising from science. This possibility will be eliminated, however, as soon as the government no longer feels reluctant to close whole areas of research as "unimportant." Any system of intellectual constructs which is espoused by the agents of secular power seems to generate its own metaphysical orthodoxy. Science fought its way free of government-sponsored religion only to threaten now to become a new government-sponsored orthodoxy.

31. According to the passage, the research trend that began during the Second World War was characterized by governmental
    (A) recognition of the importance of science to national security
    (B) support of projects that were currently non-utilitarian
    (C) inquiry into specific scientific questions
    (D) decision that certain scientific theories were dangerous
    (E) support of scientific projects that had become self-accelerating

32. The author does all of the following in his discussion EXCEPT
    (A) suggest new ways of deciding which projects should be sponsored
    (B) point out problems with present ways of deciding between projects
    (C) explain the possible consequences of present methods of evaluating specific projects
    (D) explain why governments sponsor scientific projects
    (E) explain why some projects are currently supported while others are neglected

33. It can be inferred that "Decisions based on utility" (line 25) are most probably decisions concerning projects that
    (A) attempt to answer specific questions that a government wants solved
    (B) may yield results applicable to the development of new research models
    (C) are considered laudable by the scientific community
    (D) promise to generate an elegant theory in a particular area
    (E) tend to develop a new theory to explain an aspect of the universe

34. The main purpose of the passage is to
    (A) describe the ways in which areas of scientific research have deteriorated since the Second World War
    (B) explore ways of inhibiting the growth of certain areas of scientific research
    (C) point out the areas in which new scientific theories need to be developed
    (D) examine the motives of scientists who accept government support for their research
    (E) discuss the role of supporting agencies in the differential advancement of scientific research

35. The author specifically suggests that "good" science (line 29) is equated at times with areas of research that
    (A) are financially profitable to the supporting agencies
    (B) are in a field considered crucial to immediate national interest
    (C) seem metaphysically unorthodox to those in power
    (D) may generate theories which seem elegant to those who support research
    (E) show promise of having practical applicability in the near future

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If the field of history is enormous today, that of philosophy is diffuse. In approaching the subject one must keep carefully in mind the distinction between general wisdom and the technical work of trained philosophers going about professional tasks paralleling those of economists or mathematicians. Our present interest is in philosophers in this professional sense. The century opened in the Golden Age of American philosophy when Josiah Royce, William James, George Santayana, and John Dewey were each propounding systematic thought in his own way. In this brilliant quartet, philosophy in America reached a height it has not yet regained. Despite admirable thinkers now living, it seems fair to say that no systematic philosophy propounded by any American since the death of Dewey seems likely to exercise the influence of the thinkers of the Golden Age.

But philosophers have not been idle. Stunning advances in mathematical and physical theories leading to the Einsteinian intellectual revolution of our time have given philosophy so much to analyze that it is not surprising that no general system now dominates the field. Under the merciless knife of intellectual analysis, every system of philosophy has been found inconsistent or somewhere fallacious. Because the present temper of American philosophy is extremely analytical, that sort of synthesis out of which systems develop is not at the moment much in evidence.

The matter is of more than passing importance for the humanities. In contrast to the scientist with his or her opportunity for a place in industry, and to the social scientist with his or her chance at jobs in government or business, the humanistic scholar is in the main confined to the academic world. But there are grades of opportunity even among humanists, scholars in history and literature, for example, having occasional job opportunities in libraries and special institutes. The occupational area of philosophers, however, is in the academic world, since government does not employ metaphysicians, and business and industry do not hire logicians as they hire chemists.

If the partnership of philosophy at the moment is with science in the philosophic endeavor to ascertain the basis of the dependability both of knowledge and of the intellectual process, the partnership is not irrevocable, and there is some reason to suppose there will be a slow swing back to the Platonic goal of the philosopher as citizen and magistrate.

36. The author states in the third paragraph that philosophers can be distinguished from other humanists in America in which of the following respects?

I. Their point of view
II. The areas in which they are employed
III. Their partnership with science

(A) I only
(B) II only
(C) III only
(D) I and II only
(E) II and III only

37. The author indicates that the field of philosophy in America is currently diffuse for which of the following reasons?

(A) Each person has his or her own personal philosophy.
(B) American philosophy is dominated by a single system of thought.
(C) American philosophy has never had sufficient time to formulate an identifiable system of thought.
(D) The Golden Age in America initiated four different systems of thought.
(E) American philosophy is analytical in nature.

38. Which of the following are mentioned in the passage as defects of philosophical systems?

I. Contradictions
II. Fragmented ideas
III. False ideas

(A) I only
(B) II only
(C) III only
(D) I and II only
(E) I and III only

39. According to the passage, which of the following is at present an impediment to the development of a systematic philosophy?

(A) Philosophers are currently concentrating on dissection and examination.
(B) Today's generation cannot produce professional philosophers as talented as the thinkers produced in the Golden Age.
(C) It has finally been realized that no philosophical system can be perfect.
(D) Scientific advances have relegated philosophy to a less relevant position in today's world.
(E) The tools of other disciplines, like history, are being used by philosophers.

40. The author's attitude toward philosophy's role in the future can best be described as

(A) cautiously optimistic that philosophy will become more involved in society
(B) deliberately noncommittal as to the function philosophy should perform
(C) concerned that philosophy will become merely an academic exercise
(D) regretfully resigned to the fact that the scientific disciplines must control thinking in the modern world
(E) skeptical that philosophy will attain a position equal to that held by science

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

SECTION 3 Time—30 minutes, 25 Questions

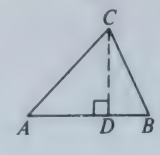
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and fill in the corresponding oval on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definition of symbols:

$=$ is equal to	$\leqq$ is less than or equal to
$\neq$ is unequal to	$\geqq$ is greater than or equal to
$<$ is less than	$\parallel$ is parallel to
$>$ is greater than	$\perp$ is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \dfrac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures that accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

1. If $x + 1 = 7$, then $(x + 2)^2 =$

(A) 25 (B) 36 (C) 49 (D) 64 (E) 81



2. In the triangle above, $p + r =$

(A) 30
(B) 45
(C) 60
(D) 90
(E) 180

3. A farmer has completed fencing all but one side of a square field. If he has already used $3x$ meters of fencing, how many meters will he need for the last side?

(A) $\frac{3x}{4}$ (B) x (C) $\frac{4x}{3}$ (D) $3x$ (E) $4x$

4. If $0 < x < \frac{1}{2}$, which of the following could NOT be x?

(A) $\frac{1}{4}$ (B) $\frac{3}{10}$ (C) $\frac{5}{8}$ (D) $\frac{7}{16}$ (E) $\frac{9}{32}$

5. There are 3 roads from Town A to Town B and 4 other roads from Town B to Town C. Mrs. Smith must drive from Town A to Town C through Town B every day to her job. How many times could Mrs. Smith make the trip from A to B to C without taking exactly the same route twice?

(A) 15 (B) 12 (C) 7 (D) 4 (E) 3

6. In a mythical country, 1 tong equals 5 sooks and 10 sooks equal 1 kant. How many tongs equal 6 kants?

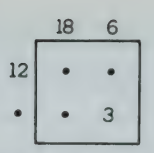
(A) $1\frac{1}{5}$ (B) $8\frac{1}{3}$ (C) 12 (D) 60 (E) $83\frac{1}{3}$



7. If the rectangle above has perimeter 20 and length 6, what is its area?

(A) 24 (B) 42 (C) 48 (D) 120
(E) It cannot be determined from the information given.

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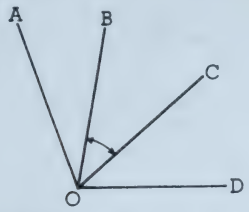
8. In the figure above, each dot represents a missing number. The product of numbers in any row or column inside the square must be equal to the number in the corresponding position outside the square. If the table is completed, which number would go below 12 ?

(A) 3 (B) 6 (C) 9 (D) 18 (E) 27

Expression 1: $6 + 2 + 1 + 3 + 4 + y$
Expression 2: $5 + 7 + 8 + y$

9. A positive number x is added to one of the two expressions above and subtracted from the other. If the resulting expressions are equal, then x =

(A) 2
(B) 3
(C) 4
(D) 5
(E) 6



10. In the figure above, $\angle AOC = 70°$, $\angle BOD = 80°$, and $\angle AOD = 110°$. What is the degree measure of $\angle BOC$?

(A) 10 (B) 20 (C) 30 (D) 40 (E) 50

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11. If $x^2 - 5x + 6 > x^2 + 5x + 6$, which of the following describes x ?

(A) $x < 0$
(B) $x = 0$
(C) $x = 2$
(D) $x = 3$
(E) $x > 3$

12. In a certain school system a child must be at least five years old on March 1 to start school on September 1 of that year. The youngest possible age, in months, on September 1, of a child starting school in this school system would be

(A) 55 (B) 56 (C) 57 (D) 65 (E) 66

13. If $\frac{4}{N}$ is an odd integer, which of the following could be a value of N ?

(A) $\frac{4}{3}$
(B) $\frac{5}{4}$
(C) $\frac{3}{4}$
(D) $\frac{2}{3}$
(E) $\frac{1}{3}$

TRI-SCHOOL MEET

	Event I	Event II	Event III
First Place (5 points)	C	A	
Second Place (3 points)	A		
Third Place (1 point)	B		

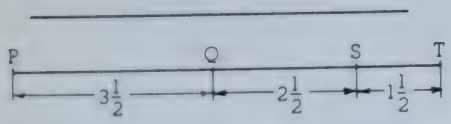
14. In a three-school meet, schools A, B, and C each entered one team for each of 3 events. If the score card above is completed and shows no ties in any event, what is the greatest possible number of points by which B's total score could exceed A's total score?

(A) 0
(B) 2
(C) 4
(D) 8
(E) 9

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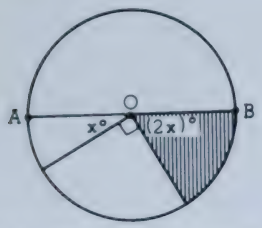
15. If $2a + 3b = 17$ and $a + 2b = 7$, then $\frac{3a + 5b}{2}$ =

(A) 10
(B) 12
(C) 18
(D) 20
(E) 24



16. On the straight line in the figure above, if M, not shown, is the midpoint of QS and if N, not shown, is the midpoint of PT, what is the length of MN ?

(A) 1 (B) $1\frac{1}{2}$ (C) 2 (D) $2\frac{1}{2}$ (E) $3\frac{3}{4}$



17. In the figure above, if the radius of the circle with diameter AB and center O is 1, then the area of the shaded region is

(A) $\frac{\pi}{12}$ (B) $\frac{\pi}{8}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{4}$ (E) $\frac{\pi}{3}$

18. If the average of p, q, and r equals the average of p and q, what is r in terms of p and q ?

(A) $\frac{p + q}{3}$ (B) $\frac{p + q}{2}$ (C) $p + q$
(D) $2(p + q)$ (E) $3(p + q)$

19. If $\frac{x + 100}{y + 200} = 1$, then y is what per cent of x ?

(A) 50% (B) 100% (C) 150% (D) 200%
(E) It cannot be determined from the information given.

20. Containers P, Q, and R have capacities in the ratio 2 to 1 to 3, respectively; P is $\frac{3}{4}$ full and Q and R are both empty. If Q is filled to capacity by pouring from P and if the remaining contents of P are poured into R, what part of R is filled?

(A) $\frac{1}{8}$
(B) $\frac{1}{6}$
(C) $\frac{1}{4}$
(D) $\frac{1}{3}$
(E) $\frac{1}{2}$

21. Five contestants competed for two days in a sports event in which there were no ties in standings. At the end of each day, the standings were recorded. Which of the following could NOT be the total number of contestants whose standings changed on the second day?

(A) One (B) Two (C) Three
(D) Four (E) Five

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22. The dimensions of two boxes are p, q, and r and s, t, and u, respectively. If $s < q$, $t < p$, $u < t$, $t < r$, and $q < u$, what is the diameter of the largest ball that will fit into both boxes?

(A) q
(B) r
(C) u
(D) t
(E) s

23. If X, Y, and Z are the hundreds', tens', and units' digits, respectively, of a number, how many hundreds are equal to that number?

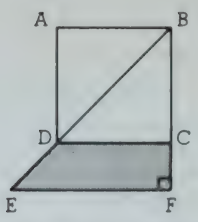
(A) $X + \frac{Y + Z}{100}$

(B) $X + \frac{10Y + Z}{100}$

(C) $\frac{X}{100} + 10Y + Z$

(D) $\frac{ZYX}{100}$

(E) $100X + 10Y + Z$



24. In the figure above, isosceles right triangle BEF overlaps square ABCD as shown. If AB = 1 and EB = 2, what is the area of the shaded region CDEF ?

(A) 2

(B) $\sqrt{2}$

(C) 1

(D) $\frac{\sqrt{2}}{2}$

(E) $\frac{1}{2}$

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25. A watch loses x minutes every y hours. At this rate, how many hours will the watch lose in 1 week

(A) 7xy (B) $\frac{7y}{x}$ (C) $\frac{x}{7y}$ (D) $\frac{14y}{5x}$ (E) $\frac{14x}{5y}$

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SECTION 5 Time—30 minutes 35 Questions

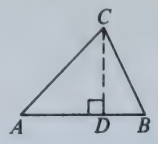
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$\neq$ is unequal to	$\geqq$ is greater than or equal to
$<$ is less than	$\parallel$ is parallel to
$>$ is greater than	$\perp$ is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.
If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

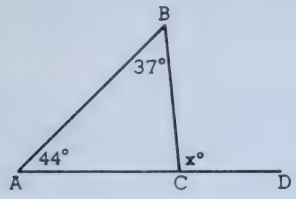
Note: Figures that accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

1. $10^3 + 10^2 + 10^1 + 1 =$

(A) 1,001
(B) 1,010
(C) 1,011
(D) 1,110
(E) 1,111

2. The list price of a certain book is \$3.95. If it is sold at a 20 per cent discount, how much does the buyer save?

(A) \$0.20 (B) \$0.79 (C) \$0.95
(D) \$1.98 (E) \$3.16



Note: Figure not drawn to scale.

3. In the figure above, if AD is a line, then $x =$

(A) 81 (B) 90 (C) 99 (D) 136 (E) 143

4. If sugar costs m cents a kilogram, how many kilograms of sugar can be bought for \$2.00 ?

(A) 2m (B) 200m (C) $\frac{2}{m}$ (D) $\frac{200}{m}$ (E) $\frac{m}{200}$

5. If $7!$ means $7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$,
$7!_e$ means $6 \cdot 4 \cdot 2$,
and $7!_o$ means $7 \cdot 5 \cdot 3 \cdot 1$,
which of the following is true?

(A) $\frac{7!_e}{7!_o} = 1$

(B) $7!_e \times 7!_o = 7!$

(C) $7!_e - 7!_o = 0$

(D) $7!_e + 7!_o = 7!$

(E) $7!_o < 7!_e$

6. If x, y, and z are consecutive integers and $x < y < z$, then, in terms of z, $x =$

(A) $z - 2$ (B) $z - 1$ (C) $z + 1$
(D) $z + 2$ (E) $\frac{z}{2}$

7. If $x = \frac{-2}{4}$, then x is equivalent to all of the following EXCEPT

(A) $\frac{-1}{2}$

(B) $-\frac{1}{-2}$

(C) $-\frac{1}{2}$

(D) $\frac{1}{-2}$

(E) $-\frac{-1}{-2}$

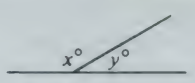
GO ON TO THE NEXT PAGE

Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet fill in oval

A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

AN E RESPONSE WILL NOT BE SCORED.

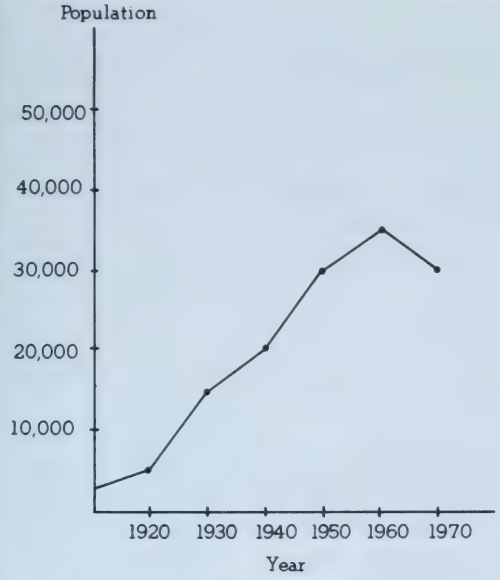
EXAMPLES

	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ Ⓔ
E2.	 $180 - x$	y	Ⓐ Ⓑ ● Ⓓ Ⓔ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ● Ⓔ

Notes:

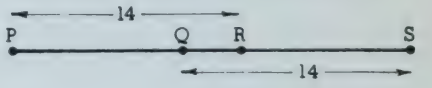
1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. In a given question, a symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

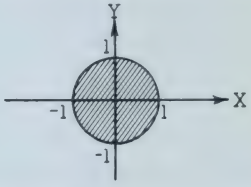
	Column A	Column B
8.	$\frac{1,058}{23}$	$\frac{1,058}{46}$



	Column A	Column B
9.	Average rate of increase in population from 1930 to 1960	Average rate of increase in population from 1940 to 1970

	Column A	Column B
10.	$\frac{1}{\frac{1}{5}}$	$\frac{1}{0.2}$

	Column A	Column B
	 QR = 3	
11.	PS	28
	$x > y + 1$	
12.	x	y

	Column A	Column B
		
13.	x if (x, y) is a point in the shaded region	0
14.	x if $2x < 25$	y if $2y > 25$
15.	Time: 12 minutes less than 3 hours	Time: $2\frac{5}{6}$ hours

GO ON TO THE NEXT PAGE

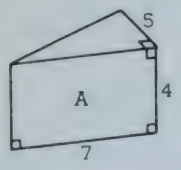
5

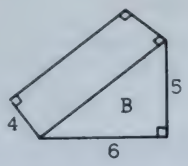
SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS

Answer: A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

AN E RESPONSE WILL NOT BE SCORED.

Column A | Column B





Solids A and B are halves of different rectangular boxes.

16. Volume of solid A | Volume of solid B

---

k and x are integers and $k^x = 64$.

17. k | x

---

18. This question was not counted in computing scores.

---

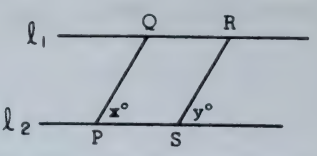
y is the least possible value of $x^2 + 1$.

19. y | 0

$\langle n \rangle = (n - 4)^2$ for all positive integers n.

20. $\langle 3 \rangle$ | $\langle 5 \rangle$

Column A | Column B



Note: Figure not drawn to scale.

Lines ℓ_1 and ℓ_2 are parallel and $x < y < 90$.

21. PQ | RS

Pat bought x items and paid 5 cents each for 3 of them and 10 cents each for the others.

22. The cost, in cents, of the x items | $15 + 10(x - 3)$

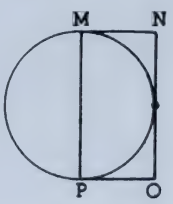
$x \neq 0$

23. $\frac{0.1 + 0.1 + 0.1}{0.1}$ | $\frac{x + x + x}{x}$

---

x and y are positive integers.

24. 90% of x plus 80% of y | 85% of (x + y)



25. Circumference of the circle with diameter MP | Perimeter of rectangle MNOP

---

26. The area of a square with perimeter 16 | The area of a parallelogram with perimeter 20

---

x is an even integer.

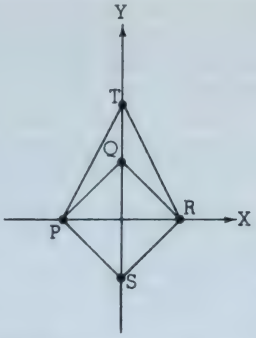
27. The number of distinct prime factors of x | The number of distinct prime factors of 2x

GO ON TO THE NEXT PAGE

Solve each of the remaining problems in this section using any available space for scratchwork. Then decide which is the best of the choices given and fill in the corresponding oval on the answer sheet.

28. A kindergarten class wants to buy a $77 tropical tree for the school. If the teacher agrees to pay twice as much as the class and the administration promises to pay 4 times as much as the class, how much should the teacher pay?

(A) $11.00 (B) $15.40 (C) $22.00
(D) $25.70 (E) $38.50



29. In square PQRS shown above, the coordinates of P and R are (−5, 0) and (5, 0) respectively. If the area of ΔPTR equals the area of square PQRS, what are the coordinates of T ?

(A) $(0, 5\sqrt{2})$ (B) (0, 10) (C) $(0, 10\sqrt{2})$
(D) (10, 0) (E) $(10\sqrt{2}, 0)$

30. If $m\left(\frac{1}{n}\right) = 1$, which of the following is NOT necessarily true?

(A) $m - n = 0$ (B) $m + n = 2n$ (C) $mn = m^2$
(D) $m^2 + n^2 = 2m^2$ (E) $m = \frac{1}{n}$

31. The average grade x of a class of 24 students was below the passing grade of 70. If 6 of the students had raised their grades by 12 points each, the average for the class would have been passing or above. Which of the following describes all possible values of x ?

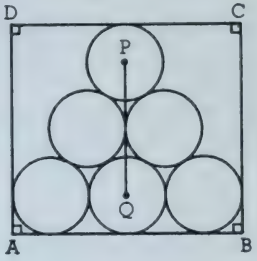
(A) $58 \leq x < 70$ (B) $67 < x < 70$ (C) $67 < x \leq 70$
(D) $67 \leq x < 70$ (E) $58 < x < 70$

32. Starting at A, Julia traveled $\frac{2}{5}$ of the distance from A to B by bus, then traveled 15 miles by car, and then walked 3 miles to reach B. For what fractional part of her trip from A to B was Julia traveling by bus or by car?

(A) $\frac{1}{5}$ (B) $\frac{3}{5}$ (C) $\frac{7}{10}$ (D) $\frac{4}{5}$ (E) $\frac{9}{10}$

33. A number is divisible by 9 if the sum of its digits is divisible by 9. Which of the following numbers is divisible by 45 ?

(A) 63,345
(B) 72,365
(C) 99,999
(D) 72,144
(E) 98,145



34. If the area of each circle in the figure above is $\frac{\pi}{4}$ and the distance between the centers of circles P and Q is $\sqrt{3}$, what is the area of rectangle ABCD ?

(A) $3 + 3\sqrt{3}$
(B) $3 + 4\sqrt{2}$
(C) 16
(D) $16 + 16\sqrt{3}$
(E) 64

35. If x, y, and z are three positive whole numbers and $x > y > z$, then, of the following, which is closest to the product xyz ?

(A) $(x - 1)yz$ (B) $x(y - 1)z$ (C) $xy(z - 1)$
(D) $x(y + 1)z$ (E) $xy(z + 1)$

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

SECTION 6 Time—30 minutes 45 Questions

For each question in this section, choose the best answer and fill in the corresponding oval on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. MERGE: (A) brighten (B) separate (C) irritate (D) harden (E) ignore

2. BEFUDDLED: (A) brawny (B) famous (C) clearheaded (D) charitable (E) energetic

3. RESIDE: (A) solicit (B) gratify (C) limit (D) measure (E) migrate

4. UNACCOUNTABLE: (A) freely given (B) widely rumored (C) slavishly obeyed (D) easily explained (E) quickly remembered

5. INCESSANT: (A) malicious (B) external (C) voluptuous (D) intermittent (E) distinctive

6. CORRUPT: (A) prove (B) punish (C) heed (D) divide (E) regenerate

7. ABRIDGE: (A) engulf (B) expand (C) tunnel (D) avoid (E) circumvent

8. TEEMING: (A) competent (B) deserted (C) watchful (D) objective (E) petulant

9. INCARCERATE: (A) purify (B) extinguish (C) disqualify (D) set free (E) speed up

10. TENACITY: (A) laxness (B) gaiety (C) graceful walk (D) happy expression (E) logical argument

11. AUSTERE: (A) indulgent (B) forthright (C) modest (D) progressive (E) mundane

12. EUPHONIOUS: (A) inappropriate (B) devious (C) raucous (D) unfriendly (E) impotent

13. GRATIS: (A) earned (B) outcast (C) broken (D) static (E) praised

14. SECULAR: (A) hypocritical (B) theoretical (C) autonomous (D) religious (E) pure

15. ENSCONCE: (A) uproot (B) measure (C) perceive (D) expedite action (E) assume authority

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that, when inserted in the sentence, best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. Truisms are beliefs that are so generally ---- that people are unaware that there could be any ---- the beliefs.

(A) contradictory. .disapproval of
(B) upheld. .substance to
(C) accepted. .opposition to
(D) supported. .reason for
(E) suspect. .questioning of

17. Although a few contemporaries ---- the book, most either ignored or mocked it.

(A) degraded (B) disregarded (C) ridiculed (D) slighted (E) appreciated

18. Lacking direct ---- racketeering, the average American citizen fails to see any reason for ----.

(A) information on. .apathy
(B) confrontation with. .alarm
(C) control of. .curiosity
(D) evidence of. .allegiance
(E) compliance with. .regulation

19. He claims that life on this planet was the result of a combination of circumstances so ---- that they could never occur again.

(A) intense (B) impressive (C) confined (D) improbable (E) unimaginable

20. Sheila Rowbotham sees her book not as an ---- work but as the initial exploration of an unmapped field.

(A) introductory (B) investigative (C) innovative (D) exhaustive (E) enlightening

GO ON TO THE NEXT PAGE

Each passage below is followed by questions based on its content. Answer the questions following each passage on the basis of what is stated or implied in that passage.

Geronimo was not the only grandchild of the Great Chief who confronted the Mexicans in hostile times. His cousin Nah-thle-tla, my mother, also encountered them during this period.

Nah-thle-tla was twelve years old when Halley's Comet appeared. She remembered "the night the stars fell." When she died at the age of one hundred and twelve, she was honored as the oldest woman in the United States. I inherited this quality from her. She also taught me all my earliest lore as my father, Nah-thle-tla's second husband, was killed in a buffalo stampede in my infancy.

Nah-thle-tla fell in love first with Shnowin, an Apache. Though his affection was intense, he was too bashful to show it and Nah-thle-tla, because of tradition, could not reveal her own feelings. Finally, Shnowin's parents visited Nah-thle-tla's family secretly to arrange the marriage price, two horses, but the young people were kept apart and the arrangements private.

After the ceremony, Shnowin went on the warpath against the Mexicans. The Apaches won a victory, but unfortunately Shnowin was killed. During his absence Nah-thle-tla had been living with her parents near the border. Apaches felt safe there since the treaty between the United States and Mexico prevented troops from crossing the border for any reason. When the Apaches saw men approaching, they believed they were the returning warriors.

Nah-thle-tla was captured by the raiders and marched south toward Chihuahua. The Mexican commander was harsh with her because of her boasted kinship with Geronimo who had, as I described, fought against the Mexicans. He made her walk to Casas Grandes, over one hundred miles, instead of riding like the other captives. He didn't realize that this was nothing unusual for an Apache woman. Nah-thle-tla never wept during this journey as did the other women. Even after they were sold into slavery she said, "We are still close to our homeland and will one day find our people again. Cheer up."

Nah-thle-tla worked for a wealthy Mexican family in Albuquerque for two years, grinding corn, making tortillas, and cooking. The señor always saw that Nah-thle-tla was supplied with the expensive fresh fruit she loved. But one night during a fiesta Nah-thle-tla escaped. She was a wiry girl, and though she was equipped with only a butcher knife, she managed to gather food, ward off animals, and walk the mountainous terrain for ten days until she reached the Magdalena Mountains. There she cried for the first time, for she saw the smoke of campfires and knew from the bands around the hair of the women that the people were Apaches.

21. The author mentions that Nah-thle-tla had fresh fruit while she lived in Albuquerque in order to give an indication of the

(A) kindness of the wealthy Mexican family
(B) sort of vegetation to be found in the region
(C) freedom Nah-thle-tla enjoyed there
(D) diet of wealthy people in the old Southwest
(E) taste for rare foods which Nah-thle-tla retained

22. What quality does the author say he inherited from his mother?

(A) Wisdom (B) Loyalty (C) Longevity
(D) Patience (E) Wiriness

23. It can be inferred that Nah-thle-tla and her parents did not flee from the Mexicans because they

(A) believed the United States Army would protect them from harm
(B) thought the Mexicans were Apache warriors
(C) realized flight was useless in their defenseless position
(D) did not see the men approaching until it was too late to flee
(E) felt sure the Mexicans would not attack women and old people

24. It can be inferred that in the paragrapns immediately preceding the passage, the author discussed

(A) Nah-thle-tla's early life with the Apaches
(B) Geronimo's battles with the Mexicans
(C) his reasons for revering Geronimo
(D) his relationship to the Great Chief
(E) Nah-thle-tla's memory of "the night the stars fell"

25. Which of the following titles best summarizes the content of the passage?

(A) Geronimo's and Nah-thle-tla's Stories
(B) An Apache Confrontation
(C) An Apache Woman in Mexico
(D) The Closeness of an Apache Family
(E) Episodes from My Mother's Past

GO ON TO THE NEXT PAGE

6

The general features of the low-level jet stream are now fairly well known. On the days that it occurs, it begins to build up in the late afternoon; it reaches its maximum in the middle of the night and decays in the early morning. At the peak of the jet, the winds in its core, between 800 and 2,000 feet up, can attain between 50 and 80 miles per hour, decreasing to 10 to 20 miles per hour between 3,000 and 4,000 feet and to zero at the ground.

Meteorologists generally agree that the diurnal cycle of heating and cooling of the earth plays a major part in the development of low-level jets. On a clear day, as the ground soaks up solar energy and grows warmer, it heats the layer of air immediately above it. If this layer gets hot enough, it begins to push its way up through the cooler air above it and a convection pattern is set up, with warm air rising over some parts of the surface and cooler air descending over others. This convective turbulence, as it is called, mixes the air at different levels. Also contributing to the mixing is mechanical turbulence, which depends on the strength of the wind and the roughness of the terrain. The interchange makes the pattern of wind speeds through the first few thousand feet more nearly uniform than it would otherwise be. Air moving closest to the surface is subject to the maximum frictional drag from the ground. As a result the wind in the lowest levels is slower than in the layers above it. Mixing partly offsets this effect: the ascending parcels of air carry up with them their lower speeds, while the descending cells bring down the higher speeds.

If the day has been calm as well as clear, with little mechanical turbulence, mixing falls off sharply as the sun goes down and the heating of the lower air decreases. The lowest air layers, still affected by surface drag but cut off now from the momentum supplied from above during the day, move more and more slowly. At the same time the upper layers are no longer sapped by contributions of momentum to the surface layers or slowed by injections of slower surface air. The winds aloft therefore speed up, and the jet begins to form.

The build-up of the jet is assisted by another nighttime weather phenomenon: temperature inversion. During the day the temperature of the first few thousand feet of atmosphere generally decreases with height. After the sun goes down, the ground begins to lose heat by radiation. If there is no blanket of clouds, the surface and the adjacent layers of air soon become cooler than the air above them. Here the temperature increases with height through the first thousand feet or so, and each succeeding parcel of air through the layer is warmer and lighter than the one below it. This is a stable arrangement, which further damps out vertical mixing. On nights when the jet develops, the depth of the inversion layer increases during the hours just after sunset, and the fastest winds are just above the top of the deepening inversion.

26. The primary purpose of the author apparently is t

(A) explain meteorological terminology
(B) illustrate what is meant by "convective turbulence"
(C) explain the effects of the diurnal cycle
(D) describe the formation of low-altitude jet streams
(E) relate temperature inversion to jet stream formation

27. According to the passage, temperature inversion is important in the formation of the low-level jet stream because it

(A) reduces mixing
(B) occurs at night
(C) cools the surface air
(D) creates layers of air
(E) reduces heat loss by radiation

28. On the basis of the passage, when does a low-level jet stream seem most likely to be at maximum intensity?

(A) Morning (B) Noon (C) Midafternoon (D) Evening (E) Midnight

29. By mixing the air from different levels, mechanic turbulence and convective turbulence tend to

(A) create winds particularly at the higher levels
(B) slow the lower-level air
(C) speed up the upper-level air
(D) produce temperature inversion
(E) equalize the momentum at the two levels

30. It can be inferred from the passage that the condition most essential for the formation of a low-level jet stream is

(A) a rise in the temperature of the higher levels of air
(B) a minimum of vertical mixing of air
(C) surface drag on the lowest air layers
(D) mechanical turbulence during the day
(E) a maximum of momentum transfer between air layers

GO ON TO THE NEXT PAGE

Select the word or set of words that best completes each of the following sentences.

31. ---- for its own sake is so admired that we find ourselves paying respectful attention not only to Channel swimmers but also to flagpole sitters or people who can dance without stopping for the longest period of time.

(A) Skill (B) Fame (C) Levity
(D) Endurance (E) Bravery

32. Accounts of scientific experiments are generally ---- because those writing about science are careful in checking the ---- of their reports.

(A) routine..reviews
(B) tentative..sources
(C) correct..accuracy
(D) misunderstood..wording
(E) misleading..validity

33. Nearly all the cultivated plants utilized by the Chinese have been of ---- origin; even rice, though known in China since Neolithic times, came from India.

(A) foreign (B) ancient (C) wild
(D) obscure (E) common

34. Loud people sometimes imply that a subdued tone is inconsistent with ----, but, unless it can be shown that the seat of candor is in the lungs, there seems to be no reason why a loud person should not be given to ----.

(A) openness..deception
(B) effectiveness..competence
(C) falsehood..hypocrisy
(D) intelligence..wisdom
(E) subtlety..straightforwardness

35. We jokingly described our doctor as a ----, for it seemed he habitually called all sorts of ---- by the same name and was thus able to prescribe the same treatment for each one.

(A) prophet..symptoms
(B) therapist..remedies
(C) specialist..maladies
(D) philanthropist..patients
(E) hypochondriac..ailments

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. CHOP : AX :: (A) knit : sweater
(B) cut : scissors (C) unwind : spool
(D) deflate : balloon (E) aim : rifle

37. ORCHESTRA : CONDUCTOR :: (A) class : student
(B) baton : majorette (C) chorus : vocalist
(D) jury : witness (E) troupe : director

38. DOODLE : DRAW :: (A) stroke : swim
(B) battle : fight (C) meander : walk
(D) flatter : praise (E) bemoan : complain

39. REDUCE : LESS :: (A) dismantle : equal
(B) conceal : valuable (C) illuminate : pale
(D) pinpoint : obvious (E) restrain : active

40. SKEPTICAL : DOUBT :: (A) shrewd : resent
(B) emotional : love (C) critical : pity
(D) credulous : trust (E) zealous : suffer

41. DESECRATION : SHRINE :: (A) deference : royalty
(B) vandalism : prisoner (C) slander : reputation
(D) pillage : war (E) blasphemy : heretic

42. OSSIFY : BONE :: (A) congeal : metal
(B) freeze : ice (C) excise : tumor
(D) stimulate : nerve (E) anoint:oil

43. DEPLORE : REGRETFULNESS :: (A) deride : contempt
(B) defy : enemy (C) differ : condescension
(D) dominate : inferiority (E) deprive : misery

44. PARRY : BLOW :: (A) kindle : fire
(B) ban : book (C) repudiate : friend
(D) denounce : traitor (E) evade : question

45. OVERWROUGHT : EXCITED ::
(A) amplified : intense (B) diversified : interesting
(C) vain : irritating (D) baffled : angry
(E) surfeited : full

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

Correct Answers for Scholastic Aptitude Test
Form Code 1Z

VERBAL		MATHEMATICAL	
Section 1	**Section 6**	**Section 3**	**Section 5**
1. C	1. B	1. D	1. E
2. E	2. C	2. D	2. B
3. D	3. E	3. B	3. A
4. D	4. D	4. C	4. D
5. C	5. D	5. B	5. B
6. D	6. E	6. C	6. A
7. E	7. B	7. A	7. B
8. A	8. B	8. C	*8. A
9. B	9. D	9. A	*9. A
10. E	10. A	10. D	*10. C
11. C	11. A	11. A	*11. B
12. B	12. C	12. E	*12. A
13. C	13. A	13. A	*13. D
14. C	14. D	14. A	*14. B
15. C	15. A	15. B	*15. B
16. B	16. C	16. A	*16. A
17. C	17. E	17. C	*17. D
18. C	18. B	18. B	*18. †
19. C	19. D	19. E	*19. A
20. E	20. D	20. B	*20. C
21. B	21. A	21. A	*21. A
22. D	22. C	22. E	*22. C
23. B	23. B	23. B	*23. C
24. A	24. B	24. E	*24. D
25. C	25. E	25. E	*25. A
26. C	26. D		*26. D
27. D	27. A		*27. C
28. E	28. E		28. C
29. A	29. E		29. B
30. B	30. B		30. E
31. B	31. D		31. D
32. A	32. C		32. E
33. A	33. A		33. E
34. E	34. A		34. A
35. D	35. C		35. A
36. B	36. B		
37. E	37. E		
38. E	38. C		
39. A	39. D		
40. A	40. D		
	41. C		
	42. B		
	43. A		
	44. E		
	45. E		

*Indicates four-choice questions. (All of the other questions are five-choice.)
†Not scored

The Scoring Process

Machine-scoring is done in three steps:

- *Scanning.* Your answer sheet is "read" by a scanning machine and the oval you filled in for each question is recorded on a computer tape.
- *Scoring.* The computer compares the oval filled in for each question with the correct response. Each correct answer receives one point; omitted questions do not count toward your score. For each wrong answer, a fraction of a point is subtracted to correct for random guessing. For questions with five answer choices, one-fourth of a point is subtracted for each wrong response; for questions with four answer choices, one-third of a point is subtracted for each wrong response. The SAT-verbal test has 85 questions with five answer choices each. If, for example, a student has 44 right, 32 wrong, and 9 omitted, the resulting raw score is determined as follows:

$$44 \text{ right} - \frac{32 \text{ wrong}}{4} = 44 - 8 = 36 \text{ raw score points}$$

Obtaining raw scores frequently involves the rounding of fractional numbers to the nearest whole number. For example, a raw score of 36.25 is rounded to 36, the nearest whole number. A raw score of 36.50 is rounded upward to 37.

- *Converting to reported scaled score.* Raw test scores are then placed on the College Board scale of 200 to 800 through a process that adjusts scores to account for minor differences in difficulty among different editions of the test. This process, known as equating, is performed so that a student's reported score is not affected by the edition of the test taken nor by the abilities of the group with whom the student takes the test. As a result of placing SAT scores on the College Board scale, scores earned by students at different times can be compared. For example, an SAT-verbal score of 400 on a test taken at one administration indicates the same level of developed verbal ability as a 400 score obtained on a different edition of the test taken at another time.

How to Score the Test

SAT-Verbal Sections 1 and 6

Step A: Count the number of correct answers for *section 1* and record the number in the space provided on the worksheet on the next page. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for *section 6* and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: To obtain C, add subtotal A to subtotal B, keeping any decimals. Enter the resulting figure on the worksheet.

Step D: To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step E: To find your reported SAT-verbal score, look up the total raw verbal score you obtained in step D in the conversion table on page 146. Enter this figure on the worksheet.

SAT-Mathematical Sections 3 and 5

Step A: Count the number of correct answers and the number of incorrect answers for *section 3* and record the numbers in the spaces provided on the worksheet. To determine the subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal C}$$

Step D: To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals. Enter the resulting figure on the worksheet.

Step E: To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step F: To find your reported SAT-mathematical score, look up the total raw mathematical score you obtained in E in the conversion table on page 146. Enter this figure on the worksheet.

SAT SCORING WORKSHEET

SAT-Verbal Sections

A. Section 1: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal A

B. Section 6: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw score (Rounded to nearest whole number) _____ D

E. SAT-verbal reported scaled score (See the conversion table on page 146.) [ ] SAT-verbal score

SAT-Mathematical Sections

A. Section 3: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal A

B. Section 5: Questions 1 through 7 and 28 through 35 (5-choice) _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal B

C. Section 5: Questions 8 through 27 (4-choice) _____ (no. correct) − ⅓ (_____ no. incorrect) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw score (Rounded to nearest whole number) _____ E

F. SAT-mathematical reported scaled score (See the conversion table on page 146.) [ ] SAT-math score

Score Conversion Table
Scholastic Aptitude Test
Form Code 1Z

Raw Score	College Board Reported Score		Raw Score	College Board Reported Score	
	SAT-Verbal	SAT-Math		SAT-Verbal	SAT-Math
85	800		40	470	630
84	790		39	460	620
83	780		38	460	610
82	770		37	450	600
81	760		36	440	590
80	750		35	430	580
79	740		34	430	570
78	740		33	420	560
77	730		32	410	560
76	720		31	410	550
75	710		30	400	540
74	710		29	390	530
73	700		28	390	520
72	690		27	380	510
71	690		26	370	500
70	680		25	360	490
69	670		24	360	480
68	670		23	350	470
67	660		22	340	470
66	650		21	340	460
65	640		20	330	450
64	640		19	320	440
63	630		18	320	430
62	620		17	310	420
61	620		16	300	410
60	610		15	290	400
59	600	800	14	290	390
58	600	790	13	280	390
57	590	780	12	270	380
56	580	770	11	270	370
55	570	760	10	260	360
54	570	750	9	250	350
53	560	740	8	240	340
52	550	730	7	240	330
51	550	730	6	230	320
50	540	720	5	220	310
49	530	710	4	220	300
48	530	700	3	210	300
47	520	690	2	200	290
46	510	680	1	200	280
45	500	670	0	200	270
44	500	660	−1	200	260
43	490	650	−2	200	250
42	480	650	−3	200	240
41	480	640	−4	200	230
			−5	200	220
			−6	200	210
			−7	200	210
			−8 or below	200	200